

AP CALCULUS AB – UNIT 3

Differentiation: Composite, Implicit, and Inverse Functions

Expanded Complete-Coverage Coloured Review



Name: _____ Class: _____ Date: _____

Review Scope and Learning Objectives

- Recognize composite structure and apply the Chain Rule in symbolic, tabular, and graphical settings.
- Differentiate implicit relations and determine tangent, normal, and higher-derivative information.
- Apply the derivative-of-an-inverse theorem from formulas, tables, and graphs.
- Differentiate inverse trigonometric functions and combine all derivative rules strategically.
- Review every idea, theorem hypothesis, representation, procedure, and common misconception appearing in the website lessons.

Complete Lesson Coverage Audit

Every website lesson is given its own mastery section below. Each section contains the complete lesson objectives, strategic reminders, a solved model, and foundation-to-challenge practice. The final synthesis sections mix lessons so that students must select procedures rather than follow a stated template.

Complete Website Lesson Coverage Map

Lesson	Website title	Ideas explicitly reviewed
3.1	The Chain Rule	Recognize composite functions and identify their inner and outer functions.; Differentiate composite functions using the Chain Rule in prime and Leibniz notation.; Combine the Chain Rule with power, trigonometric, exponential, logarithmic, product, and quotient rules.; Use tables, graphs, and contextual information to evaluate derivatives of compositions.
3.2	Implicit Differentiation	Differentiate relations in which y is defined implicitly as a function of x .; Solve algebraically for dy/dx and evaluate it at points on an implicit curve.; Determine tangent and normal lines, horizontal and vertical tangents, and second derivatives.; Recognize when a derivative formula is inconclusive at a singular point.
3.3	Inverse Function and Inverse Trigonometric Derivatives	Use the derivative-of-an-inverse formula from equations, tables, and graphs.; Differentiate inverse trigonometric functions and compositions containing them.; Write tangent lines to inverse functions and interpret reciprocal slopes.; Combine inverse derivatives with the Chain, Product, and Quotient Rules.

Unit Knowledge, Definitions, and Theorem Library

Key Knowledge, Formulas, and Theorems

- Chain Rule: $\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)$. Apply one inner derivative factor for every layer.
- Composition order matters: $(f \circ g)'$ and $(g \circ f)'$ generally differ.
- Implicit differentiation treats y as a function of x , so differentiating a term involving y introduces dy/dx .

- After implicit differentiation, collect all y' terms, factor, and solve; verify the point lies on the original curve.
- Horizontal tangents usually satisfy the derivative numerator = 0 with denominator $\neq 0$; vertical tangents reverse these roles. A $0/0$ derivative formula is inconclusive.
- Higher implicit derivatives require differentiating the first-derivative relation again and substituting known values carefully.
- Inverse derivative: $(f^{-1})'(b) = \frac{1}{f'(a)}$ where $f(a) = b$ and $f'(a) \neq 0$. Evaluate at the preimage.
- Inverse graphs reflect across $y = x$, exchanging coordinates and reciprocal nonzero slopes.
- Inverse trigonometric derivatives must be combined with the Chain Rule; domain restrictions and radians matter.
- Advanced differentiation often combines Chain, Product, Quotient, implicit, and inverse procedures in one expression.

Strategy and Precision Checklist

1. Name the representation and the requested quantity before calculating.
2. State all hypotheses before using a theorem.
3. Preserve exact values in non-calculator work and use radians for trigonometric calculus.
4. A graph or table supports an estimate; an analytic proof requires a definition, theorem, or valid algebraic argument.
5. Contextual conclusions must include the instant, sign, units, and meaning.
6. Candidate conditions such as a zero derivative, zero second derivative, or zero denominator require sign/domain verification before a conclusion.

Lesson 3.1: The Chain Rule**Complete objectives**

- Recognize composite functions and identify their inner and outer functions.
- Differentiate composite functions using the Chain Rule in prime and Leibniz notation.
- Combine the Chain Rule with power, trigonometric, exponential, logarithmic, product, and quotient rules.
- Use tables, graphs, and contextual information to evaluate derivatives of compositions.

Strategic reminders

1. Mark the innermost expression before differentiating. Rewrite radicals and reciprocals as powers when useful.
2. Differentiate the outside while leaving the inside unchanged, then multiply by the derivative of the inside.
3. For several layers, repeat the process until the derivative reaches the original independent variable.
4. When product or quotient structure is present, preserve that structure and apply the Chain Rule inside its factors.
5. Do not write $f'(x)g'(x)$ for the derivative of $f(g(x))$; the outer derivative is evaluated at $g(x)$.

Solved Example: Power outside

Differentiate $y = (3x^2 - 5x + 1)^4$.

Answer and Reasoning

Let $u = 3x^2 - 5x + 1$. Then $\frac{dy}{dx} = 4u^3 \frac{du}{dx} = 4(3x^2 - 5x + 1)^3(6x - 5)$.

Lesson Mastery Practice

1. Identify inner and outer functions for $y = \sqrt{5x^2 + 1}$.

Foundation

2. Differentiate $y = e^{x^2 - 3x}$.

Core

3. Differentiate $y = x^3 \cos(x^2)$.

Advanced

4. If differentiable f is even, prove that f' is odd.

Challenge

AP-Style Reasoning Check

If $y = (1 - 3x^2)^5$, then y' is

(A) $5(1 - 3x^2)^4$

(C) $-6x(1 - 3x^2)^5$

(B) $-30x(1 - 3x^2)^4$

(D) $5(-6x)^4$

Lesson 3.2: Implicit Differentiation**Complete objectives**

- Differentiate relations in which y is defined implicitly as a function of x .
- Solve algebraically for dy/dx and evaluate it at points on an implicit curve.
- Determine tangent and normal lines, horizontal and vertical tangents, and second derivatives.
- Recognize when a derivative formula is inconclusive at a singular point.

Strategic reminders

1. Differentiate every term with respect to x and attach y' whenever the differentiated expression contains y .
2. Collect all terms containing y' on one side before factoring and dividing.
3. Verify that a stated point lies on the original curve before using it.
4. For a normal line, use the negative reciprocal of a finite nonzero tangent slope.
5. If both numerator and denominator of the derivative formula vanish, return to the original relation; the formula is inconclusive.

Solved Example: Circle and tangent line

For $x^2 + y^2 = 25$, find y' and the tangent line at $(3, 4)$.

Answer and Reasoning

$2x + 2yy' = 0$, so $y' = -x/y$. At $(3, 4)$, $m = -3/4$ and $y - 4 = -\frac{3}{4}(x - 3)$.

Lesson Mastery Practice

5. For $x + y = 7$, find y' .

Foundation

6. Write the tangent line to $x^2 + xy = 6$ at $(2, 1)$.

Core

7. For $x^3 + y^3 = 6xy$, find y' .

Advanced

8. At $(0, 0)$ on $y^2 = x^2(x + 1)$, use leading terms to identify candidate tangent slopes.

Challenge

AP-Style Reasoning Check

If $x^2 + y^2 = 9$, then y' equals

(A) $-x/y$

(B) $-y/x$

(C) x/y

(D) y/x

Lesson 3.3: Inverse Function and Inverse Trigonometric Derivatives**Complete objectives**

- Use the derivative-of-an-inverse formula from equations, tables, and graphs.
- Differentiate inverse trigonometric functions and compositions containing them.
- Write tangent lines to inverse functions and interpret reciprocal slopes.
- Combine inverse derivatives with the Chain, Product, and Quotient Rules.

Strategic reminders

1. To find $(f^{-1})'(b)$, first identify the preimage a satisfying $f(a) = b$.
2. Evaluate f' at the preimage, not at the output b , then take the reciprocal.
3. For inverse trigonometric compositions, multiply the standard derivative by the derivative of the inside.
4. In a table, locate the output in the $f(x)$ row before reading the corresponding derivative entry.
5. Use radians for trigonometric derivative formulas.

Solved Example: Inverse derivative from a formula

Let $f(x) = x^3 + x$. Find $(f^{-1})'(2)$.

Answer and Reasoning

Since $f(1) = 2$ and $f'(1) = 4$, $(f^{-1})'(2) = 1/4$.

Lesson Mastery Practice

9. If $f(2) = 7$ and $f'(2) = 5$, find $(f^{-1})'(7)$.

Foundation

10. Let $f(x) = x^3 + 2x$. Find $(f^{-1})'(3)$.

Core

11. Differentiate $y = \arcsin\left(\frac{x}{1+x}\right)$.

Advanced

12. Prove the inverse-derivative formula from $f(f^{-1}(x)) = x$.

Challenge

AP-Style Reasoning Check

If $f(4) = 9$ and $f'(4) = -3$, then $(f^{-1})'(9)$ is

(A) -3

(B) $-1/3$

(C) $1/3$

(D) 3

Unit 3 Difficult and Challenge Synthesis

1. Differentiate $y = \left[1 + \ln(1 + e^{x^2})\right]^5$ and display every Chain Rule layer.

Difficult

2. For $x^2 + xy + y^2 = 7$, find all points where the tangent is horizontal and all points where it is vertical.

Challenge

3. If $f(2) = 5$, $f'(2) = -3$, and $f''(2) = 4$, find $(f^{-1})'(5)$ and $(f^{-1})''(5)$.

Challenge

4. Find dy/dx and d^2y/dx^2 at $(3, 4)$ on $x^2 + y^2 = 25$.

Difficult

5. Differentiate $F(x) = x^2 \arctan\left(\frac{1-x}{1+x}\right)$ on its domain.

Challenge

6. An implicit derivative simplifies to $y' = \frac{x-y}{x+y}$, but at $(0, 0)$ it gives $0/0$. Explain why the formula is inconclusive and describe what must be done.

Hard

Cumulative AP-Style Checkpoint

1. If $F(x) = \sin((1 + x^2)^4)$, then $F'(x)$ is

- (A) $4(1 + x^2)^3 \cos x$ (B) $8x(1 + x^2)^3 \sin((1 + x^2)^4)$
 (C) $\cos(4(1 + x^2)^3)$ (D) $8x(1 + x^2)^3 \cos((1 + x^2)^4)$

2. For $x^2 + xy + y^2 = 7$, dy/dx is

- (A) $-\frac{2x+y}{x+2y}$ (B) $-\frac{x+2y}{2x+y}$
 (C) $-\frac{2x}{2y}$ (D) $\frac{2x+y}{x+2y}$

3. If $f(3) = 8$ and $f'(3) = -4$, then $(f^{-1})'(8)$ is

- (A) $\frac{1}{8}$ (B) $-\frac{1}{4}$
 (C) -4 (D) $\frac{1}{4}$

4. Given $g(1) = 2$, $g'(1) = -3$, and $f'(2) = 5$, $(f \circ g)'(1)$ equals

- (A) 8 (B) -15
 (C) 10 (D) 2

5. An isolated formula for y' gives $0/0$ at a point on an implicit curve. The best conclusion is

- (A) The tangent is automatically vertical (B) The point cannot lie on the curve
 (C) The derivative formula is inconclusive and the original relation must be analyzed (D) The tangent is automatically horizontal

6. $\frac{d}{dx} \arcsin(2x)$ equals

- (A) $\frac{1}{\sqrt{1-4x^2}}$ (B) $\frac{2}{1+4x^2}$
 (C) $\frac{1}{2\sqrt{1-x^2}}$ (D) $\frac{2}{\sqrt{1-4x^2}}$

7. Differentiate $x(x^2 + 3)^2$.

- (A) $2x(x^2 + 3)$ (B) $(x^2 + 3)^2$
 (C) $4x(x^2 + 3)$ (D) $(x^2 + 3)(5x^2 + 3)$

8. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

- (A) $-\frac{x+2y}{2x+y}$ (B) $-\frac{2x+1}{1+2y}$
 (C) $\frac{2x+y}{x+2y}$ (D) $-\frac{2x+y}{x+2y}$

9. If $f(4) = 3$ and $f'(4) = 2$, find $(f^{-1})'(3)$.

- (A) 2 (B) $-\frac{1}{2}$
 (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

10. Differentiate $x(x^2 + 3)^4$.

- (A) $(x^2 + 3)^4$ (B) $8x(x^2 + 3)^3$
 (C) $(x^2 + 3)^3(9x^2 + 3)$ (D) $4x(x^2 + 3)^3$

11. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

- (A) $-\frac{2x+y}{x+2y}$ (B) $-\frac{x+2y}{2x+y}$
(C) $\frac{2x+y}{x+2y}$ (D) $-\frac{2x+1}{1+2y}$

12. Differentiate $\arcsin(3x)$.

- (A) $\frac{3}{1+9x^2}$ (B) $\frac{3}{\sqrt{1-9x^2}}$
(C) $\frac{1}{\sqrt{1-9x^2}}$ (D) $-\frac{3}{\sqrt{1-9x^2}}$

13. Find $\frac{d}{dx}[\sin(3x^2)]$.

- (A) $6x \sin(3x^2)$ (B) $\cos(3x^2)$
(C) $3 \cos(3x^2)$ (D) $6x \cos(3x^2)$

14. For the implicit curve $x^2 + xy + y^2 = K$, which formula for dy/dx is correct?

- (A) $\frac{2x+y}{x+2y}$ (B) $-\frac{x+2y}{2x+y}$
(C) $-\frac{2x+1}{1+2y}$ (D) $-\frac{2x+y}{x+2y}$

15. Differentiate $\arctan(x^2)$.

- (A) $\frac{1}{1+x^4}$ (B) $\frac{2x^1}{\sqrt{1-x^4}}$
(C) $\frac{2x^1}{1+x^2}$ (D) $\frac{2x}{x^4+1}$

16. Differentiate e^{4x^3} .

- (A) $12x^2e^{4x}$ (B) $12x^2$
(C) e^{4x^3} (D) $12x^2e^{4x^3}$

17. For $x^2 + y^2 = 13$, find the slope at $(2, 3)$.

- (A) $-\frac{3}{2}$ (B) $-\frac{2}{3}$
(C) $\frac{3}{2}$ (D) $\frac{2}{3}$

18. If $f(3) = 1$ and $f'(3) = 2$, find $(f^{-1})'(1)$.

- (A) 2 (B) $\frac{1}{3}$
(C) $-\frac{1}{2}$ (D) $\frac{1}{2}$

19. Differentiate $\ln(x^2 + 1)$.

- (A) $2x \log(x^2 + 1)$ (B) $\frac{1}{x^2+1}$
(C) $\frac{2}{x^2+1}$ (D) $\frac{2x}{x^2+1}$

20. For $x^2 + y^2 = 25$, find the slope at $(4, 3)$.

- (A) $-\frac{3}{4}$ (B) $-\frac{4}{3}$
(C) $\frac{3}{4}$ (D) $\frac{4}{3}$

Cumulative Free-Response Synthesis

FRQ1 – Composition and inverse data

	x	1	2	4	7
	$f(x)$	4	7	1	2
The table gives	$f'(x)$	-3	5	2	-1
	$g(x)$	2	4	7	1
	$g'(x)$	6	-2	3	4

. Assume f is one-to-one.(a) Find $(f \circ g)'(1)$ and $(g \circ f)'(1)$.

[4 points]

(b) Find $(f^{-1})'(7)$.

[2 points]

(c) Find the tangent line to $y = f^{-1}(x)$ at $x = 7$.

[2 points]

FRQ2 – Implicit curve and tangent classification

The curve $x^2 + xy + y^2 = 7$ is considered.(a) Find dy/dx .

[3 points]

(b) Find all points with horizontal tangents.

[3 points]

(c) Find all points with vertical tangents.

[3 points]

FRQ3 – Multi-layer differentiation

$$\text{Let } F(x) = \left[1 + \ln \left(1 + e^{x^2} \right) \right]^5.$$

(a) Identify the successive inner functions from the deepest layer outward. [2 points]

(b) Find $F'(x)$. [4 points]

(c) Explain why $F'(0) = 0$ without evaluating all other factors numerically. [2 points]

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